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Automatic infusion valve

Abstract

The present disclosure generally relates to fluid control valves for delivering and/or aspirating fluid during ophthalmic surgeries and procedures. In one embodiment, a valve assembly includes a first portion configured to fluidly couple with a gas supply line and a second portion configured to fluidly couple with a liquid supply line and an infusion line. The first portion and the second portion are partitioned or separated from each other by a filter having a hydrophobic membrane configured to prevent the flow of liquids therethrough while allowing the free flow of gas. Accordingly, an infusion liquid may be flown through the second portion while gases may be simultaneously aspirated through the first portion, without any liquids travelling into the gas supply line. The gas supply line may thus be utilized to vent or purge gases from the infusion line before or during performance of surgical procedures.

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Background/Summary

PRIORITY CLAIM (1) This application claims the benefit of priority of U.S. Provisional Patent Application Ser. No. 63/126,675 titled “AUTOMATIC INFUSION VALVE,” filed on Dec. 17, 2020, whose inventors are Sean Christopher Madden and Parthasarathy Parishram, which is hereby incorporated by reference in its entirety as though fully and completely set forth herein.

BACKGROUND

Field

(1) Embodiments of the present disclosure generally relate to devices for ophthalmic procedures, and more particularly, to devices for intraocular fluid delivery.

Description of the Related Art

(2) Microsurgical procedures frequently require precision cutting, removal, and manipulation of various body tissues. For example, certain ophthalmic surgical procedures, such as pars plana vitrectomies, require cutting and removal of portions of the vitreous humor, a transparent gel-like material that fills the posterior segment of the eye. Simultaneously while removing the vitreous humor, a liquid solution (e.g., balanced salt solution (BSS)) is typically infused into the eye to maintain intraocular pressure and prevent collapse of the eye wall. In cases of retinal breaks or retinal detachment, the liquid solution may then be exchanged for air, through a process known as fluid-air exchange, to help push out subretinal fluid from the intraocular space while maintaining intraocular pressure and temporarily holding the retina in place. During such procedures, the liquid and air are provided by separate supply lines that are conjoined with a singular downstream infusion line via a stopcock.

(3) In some cases, the air pressure in the gas supply line may build up and cause air to escape into the infusion line, forming air bubbles in the infusion liquid which may travel to the eye and negatively affect intraocular pressure during surgery. Conventional designs for check valves of infusion stopcocks, however, do not allow venting through the gas supply line without reverse leakage of liquid and thus, there is currently no effective way to remove the air bubbles or prevent

their escape into infusion liquids. Additionally, during some procedures, infusion fluids must be back-flowed through the infusion line in order for other surgical fluids, such as a retinal tamponade, to be injected into the intraocular space. In such cases, the amount of infusion fluid that may be back-flowed is limited by the inability to purge infusion gases through the gas supply line without reverse leakage of liquids therein, which may damage the air pump and/or cause additional complications during fluid infusion.

(4) Therefore, what is needed in the art are improved fluid control valves for ophthalmic fluid infusion that enable aspiration and purging of gases.

SUMMARY

(5) The present disclosure generally relates to devices for surgical procedures, and more particularly, surgical devices for ophthalmic fluid infusion and aspiration.

(6) In certain embodiments, a valve assembly for fluid infusion is provided. The valve assembly includes a first portion with a first conduit having a first port and a first cavity having a proximal end and a distal end opposite of the proximal end. The proximal end is fluidly coupled to the first port via the first conduit, and the distal end has a cross-sectional area greater than a cross-sectional area of the first conduit. The valve assembly further includes a second portion with a second conduit having a second port, a third conduit having a third port, and a second cavity disposed between the second conduit and the third conduit and fluidly coupling the second port to the third port, where the second cavity is adjacent to the first cavity. A filter having a hydrophobic membrane is disposed between and partitions the first cavity from the second cavity and the hydrophobic membrane partially defines the second cavity.

(7) In certain embodiments, a fluid infusion system for ophthalmic procedures is provided. The fluid infusion system includes a surgical console having a first fluid line coupled to a gas fluid source and a second fluid line coupled to a liquid fluid source. The fluid infusion system further includes a valve assembly fluidly coupled to the first fluid line and the second fluid line. The valve assembly includes a first conduit having a first port in fluid communication with the first fluid line, a second conduit having a second port in fluid communication with the second fluid line, and a third conduit having a third port in fluid communication with a third fluid line. Flow rates of fluids through the first, second, and third lines is controlled by the surgical console. The first, second, and third conduits are further coupled to an intermediary cavity and in fluid communication with each other. A filter is disposed within the intermediary cavity and partitions the first conduit from the second and third conduits. The filter includes a hydrophobic membrane disposed on a side thereof opposite the first conduit and configured to prevent flow of liquids from the second and third conduits into the first conduit while allowing bi-directional flow of gases therebetween.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above-recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, and may admit to other equally effective embodiments.

(2) FIG. 1 illustrates a perspective view of an exemplary surgical console, according to certain embodiments of the present disclosure.

(3) FIG. 2A illustrates a perspective view of an exemplary valve assembly, according to certain embodiments of the present disclosure.

(4) FIG. 2B illustrates a perspective exploded view of the valve assembly of FIG. 2A, according to

certain embodiments of the present disclosure.

(5) FIG. 2C illustrates a schematic cross-sectional view of the valve assembly of FIG. 2A, according to certain embodiments of the present disclosure.

(6) FIG. 3A illustrates a perspective view of an exemplary valve assembly, according to certain embodiments of the present disclosure.

(7) FIG. 3B illustrates a perspective exploded view of the valve assembly of FIG. 3A, according to certain embodiments of the present disclosure.

(8) FIG. 3C illustrates a schematic cross-sectional view of the valve assembly of FIG. 3A, according to certain embodiments of the present disclosure.

(9) FIG. 4A illustrates a schematic plan view of an exemplary operational mode of the valve assemblies of FIGS. 2A-2C and 3A-3C, according to certain embodiments of the present disclosure.

(10) FIG. 4B illustrates a schematic plan view of an exemplary operational mode of the valve assemblies of FIGS. 2A-2C and 3A-3C, according to certain embodiments of the present disclosure.

(11) FIG. 4C illustrates a schematic plan view of an operational mode of the valve assemblies of FIGS. 2A-2C and 3A-3C, according to certain embodiments of the present disclosure.

(12) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

(13) The present disclosure generally relates to fluid control valves for delivering and/or aspirating fluid during ophthalmic surgeries and procedures. For example, the fluid control valves described herein may be used during vitrectomies, such as pars plana vitrectomies for the treatment of posterior segment diseases. Vitrectomies typically require cutting and removal of portions of the vitreous humor. In order to maintain intraocular pressure and prevent collapse of the eye during such surgical procedures, liquid is infused into the intraocular space and thereafter aspirated. In certain procedures, the liquid is then exchanged with air or other gases during a process known as fluid-air exchange. During such processes, it is typically beneficial to purge or vent any undesired gases in the infusion line and/or the intraocular space to maintain intraocular pressure. The fluid control valves and methods described herein provide improved structures and mechanisms for infusion fluid flow regulation that enable upstream purging and/or venting of gases from infusion lines while also preventing liquids from the infusion lines to leak into the gas supply lines.

(14) In certain embodiments, a valve assembly includes a first portion configured to fluidly couple with a gas supply line and a second portion configured to fluidly couple with a liquid supply line and an infusion line. The first portion and the second portion are partitioned or separated from each other by a filter having a hydrophobic membrane configured to prevent the flow of liquids therethrough while allowing the free flow of gas. Accordingly, an infusion liquid may be flowed through the second portion while gases may be simultaneously aspirated into the first portion, without any liquids travelling into the gas supply line. The gas supply line may thus be utilized to vent or purge gases from the infusion line before or during performance of surgical procedures.

(15) FIG. 1 illustrates a perspective view of an exemplary surgical console **100** that may be utilized in combination with the fluid control valves described herein. The surgical console **100** is operably coupled, physically or wirelessly, to any number of user interfaces, including a foot controller **102** and a surgical tool **104** such as a vitrector. The surgical console **100** provides one or more port connectors **106** for physically coupling the user interfaces to various components of the surgical console **100**. For example, the surgical tool **104** may be fluidly coupled with a vacuum source via a vacuum supply line **112** disposed through a port connector **106** to enable aspiration of cut vitreous from the patient's eye. Similarly, one or more port connectors **106** may be utilized to couple a fluid

infusion system **110** with one or more infusion fluid sources, (e.g., an air/gas source, a liquid perfluorocarbon source, a silicone oil infusion (SOI) source, a BSS source, etc.) to enable infusion of fluids into the eye during vitreous removal. As shown in FIG. **1**, the fluid infusion system **110** includes an infusion line **108** fluidly coupled with a gas supply line **114** and a separate liquid supply line **116** at a three-way automatic valve assembly **118**, which may enable selective flow of different infusion fluids through the infusion line **108**.

(16) In operation, the user may control an aspect or mechanism of the surgical tool **104** and/or the fluid infusion system **110** via actuation of the foot controller **102**, which may include a foot pedal. For example, the user may press down on (e.g., depress) the foot controller **102** to initiate and increase a flow rate of an infusion fluid from a fluid source through the fluid infusion system **110** and into the eye of the patient. Alternatively, reducing depression of the foot controller **102** (e.g., lifting the user's foot) may decrease and ultimately stop the flow of fluid through the fluid infusion system **110**. Accordingly, in certain embodiments, the flow rate of infusion fluids through the fluid infusion system **110** corresponds to the amount of depression of the foot controller **102**. In certain embodiments, the surgical console **100** further includes a display **120** for displaying information to the user (the display may also incorporate a touchscreen for receiving user input). Thus, the display **120** may display information about infusion fluid parameters, such as infusion fluid flow rates and intraocular pressure, to the user during operation thereof.

(17) FIGS. **2A-2C** illustrate a valve assembly **200** for flow control of infusion fluids during surgical procedures. Valve assembly **200** is an example of the automatic three-way valve assembly **118**, which may be utilized in combination with the fluid infusion system **110** and the surgical console **100** described above. As shown in more detail in FIGS. **2A-2C**, the valve assembly **200** generally includes a hydrophobic filter (shown as hydrophobic filter **222** in FIGS. **2B-2C**) disposed between valve assembly **200**'s first portion (e.g., an upper body), configured to fluidly couple with a gas supply line, and valve assembly **200**'s second portion (e.g., a lower body), configured to fluidly couple with a fluid supply line. The partitioning of the first portion and the second portion by the hydrophobic filter enables active bi-directional flow of gases, such as air, between the gas supply line and an infusion line while passively preventing liquids from travelling into the gas supply line. Thus, gases may be vented or purged from the fluid infusion system **110** during fluid infusion to enable improved control of intraocular pressure during surgical procedures.

(18) FIG. **2A** illustrates a perspective view of the valve assembly **200**, while FIG. **2B** illustrates a perspective exploded view thereof and FIG. **2C** illustrates a cross-sectional view thereof. Accordingly, FIGS. **2A-2C** are herein described together for clarity.

(19) As noted above, the valve assembly **200** generally includes an upper body **232** configured to interface (e.g., couple) with a lower body **202**. In certain embodiments, the upper body **232** and lower body **202** are formed of any suitable plastic materials, such as acrylonitrile butadiene styrene (ABS), polycarbonate (PC), nylon, and acrylic, which may be transparent or opaque in color. The upper body **232** has a base **238** from which an arm **234** extends in a proximal direction (e.g., toward a surgical console or gas source) for coupling with gas supply line **114**. Note that, as described herein, a distal end or portion of a component refers to the end or portion that is closer in line to a patient's body during use thereof. On the other hand, a proximal end or portion of the component refers to the end or the portion that is distanced further away in line from the patient's body (e.g., closer to the surgical console).

(20) The gas supply line **114** couples with a port **237** at a proximal end of the arm **234**, which provides fluid connection with a conduit **236** extending through a length of the arm **234**. In certain embodiments, a diameter of the port **237** is substantially the same or slightly larger than an outer diameter of the gas supply line **114** to allow a distal end of the gas supply line **114** to be securely fit within the port **237**. In certain embodiments, the outer diameter of the proximal end of the arm **234** is substantially the same or slightly smaller than an inner diameter of the gas supply line **114** to allow the distal end of the gas supply line **114** to secure fit over the proximal end of the arm **234**.

(21) The conduit **236** extends from the proximal end of the arm **234** to a distal end of the arm **234** and opens into a cavity **240** within the base **238** of the upper body **232**. In certain embodiments, the arm **234** and thus the conduit **236** have one or more curved portions to create an angled flow path for gases between the gas supply line **114** and the cavity **240**. For example, the arm **234** and the conduit **236** may have a bend disposed at about a 90-degree angle between the proximal and distal ends thereof, thus creating an elbow-like gas flow path. The bending of the gas flow path enables a three-way connection of the valve assembly **200** between the gas supply line **114**, the liquid supply line **116**, and an infusion line **108**.

(22) The cavity **240** is disposed at the distal end of conduit **236** and generally has one or more dimensions greater than a width or diameter of the conduit **236**. In certain embodiments, the cavity **240** has a cross-sectional area that gradually increases from an end of the cavity **240** nearest the arm **234** to an end of the cavity **240** furthest from the arm **234** (e.g., nearest the lower body **202**). For example, the end of the cavity **240** nearest the arm **234** may have substantially the same cross-sectional area as the distal end of the conduit **236** while the end of the cavity **240** furthest from the arm **234** may have a cross-sectional area greater than a cross-sectional area of the distal end of the conduit **236**. In certain embodiments, the cavity **240** has a frustoconical-like shape. The increased cross-sectional area of the cavity **240** at the distal end thereof enables utilization of a larger-area filter **222** between the upper body **232** and the lower body **202** and provides more surface area through which gases may be vented or purged from liquids flowing between the liquid supply line **116** and the infusion line **108**.

(23) Generally, a lower surface of the upper body **232** and an upper surface of the lower body **202** are configured to interface with or engage each other and secure the filter **222** therebetween. In certain embodiments, the lower body **202** couples to the upper body **232** at a lower surface of the base **238** such that the cavity **240** faces a chamber **204** (e.g., a second cavity or reservoir) located at a central position of the lower body **202**. The chamber **204** may have a cross-sectional area substantially the same or greater than the cross-sectional area of the end of the cavity **240** nearest the chamber **204** so as not to constrict air flow between the upper body **232** and the lower body **202** and vice versa. The lower body **202** further includes extensions **206** and **210** on opposing sides of the chamber **204**, where each extension **206** and **210** has a conduit **208** or **212** formed therethrough, respectively. The conduits **208** and **212** extend from the chamber **204** in opposing directions toward ports **209** and **211** located at the proximal and distal ends of the extensions **206** and **210**, respectively. In certain embodiments, the port **209** is configured to fluidly couple with a liquid supply line **116**, while the port **211** is configured to couple to infusion line **108**.

(24) Similar to the port **237**, the port **209** may have a diameter substantially the same or slightly larger than an outer diameter of the liquid supply line **116** to allow a distal end of the liquid supply line **116** to be securely fit within the port **209**. Alternatively, an outer diameter of the proximal end of the extension **206** may be substantially the same or slightly smaller than an inner diameter of the liquid supply line **116** to allow the distal end of the liquid supply line **116** to securely fit over the proximal end of the extension **206**. The extension **212**, however, is generally sized to have an outer diameter substantially the same or slightly smaller than an inner diameter of the proximal end of the infusion line **108**. Accordingly, the infusion line **108** is configured to securely fit around the extension **212**. In certain embodiments, the extension **212** may include a locking mechanism, such as a Luer lock **250**, which is configured to couple with the infusion line **108** and provide additional mechanical holding force for a leak-free seal between the valve assembly **200** and the infusion line **108**. For example, the Luer lock **250** may comprise a threaded interior surface **252** through which the proximal end of the infusion line **108** may be secured within.

(25) The filter **222** is disposed between the cavity **240** of the upper body **232** and the chamber **204** of the lower body **202**, thus partially defining both the cavity **240** and the chamber **204**. The filter **222** includes any suitable type of membrane filter having a hydrophobic membrane **224** permeable to gas. The hydrophobic membrane **224** may also be capable of capturing individual viruses and

bacteria, thus acting as a sterile barrier to prevent viruses and bacteria from entering the eye from the low pressure gas source.

(26) In some examples, the filter **222** includes a membrane **224** formed of polytetrafluoroethylene (PTFE), expanded PTFE (ePTFE), polycarbonate track etch (PCTE), polyesters (e.g., polyethylene terephthalate (PET)), nylon, cellulose (e.g., surfactant free cellulose acetate (SCFA), cellulose nitrate (CN), cellulose acetate (CA), polyethersulfone (PES), glass fibers, or acrylic copolymers. The membrane **224** may further be unsupported or supported by a backing **226** formed of materials including but not limited to polyester, polyethylene, polypropylene, or nylon. For example, in certain embodiments, the filter **222** includes an ePTFE membrane **224** having a polyester backing **226**. Generally, the hydrophobic membrane **224** of the filter **222** is oriented to face the chamber **204** so as to prevent liquids from flowing from the chamber **204** through the filter **222** and into the cavity **240**. The membrane **224** has a pore size ranging between about 0.1 μm to about 10.0 μm , such as between about 0.2 μm to about 5 μm , such as between about 0.5 μm to about 3.0 μm , such as a between about 0.8 μm to about 1.2 μm . Furthermore, the membrane **224** may have a thickness ranging between about 150 μm to about 300 μm , such as between about 200 μm to about 250 μm .

(27) During operation, infusion liquid from the liquid source, such as silicone oil or balanced salt solution (BSS), may flow through the liquid supply line **116**, into the lower body **202** of the valve assembly **200**, and through the infusion line **108** toward the patient's eye and vice-versa.

Alternatively, infusion gases from the gas source, such as air, may flow through the gas supply line **114**, into the upper body **232**, past the filter **222** into the lower body **202**, and then into the infusion line **108** toward the patient's eye and vice-versa. The placement of the hydrophobic filter **222** between the upper body **232** and the lower body **202** passively prevents liquid from flowing up into the upper body **232** and the gas supply line **114**, while allowing gases to pass therethrough.

Accordingly, the valve assembly **200** enables the venting, purging, and/or back-flow of gases during fluid infusion procedures while preventing the escape of liquid into the gas supply line **116**, which is described in further detail below.

(28) Please note that although a single filter **222** is depicted in FIGS. 2A-2C, in certain embodiments, it is further contemplated that the valve assembly **200** may include two or more filters arranged in a linear or stacked configuration. The two or more filters may be formed of different materials and/or have different pore sizes relative to each other. For example, in certain embodiments, the valve assembly **200** may include a second filter disposed between the upper and lower bodies **232**, **202** and upstream of the filter **222** (e.g., closer in line to the gas supply line **114**). In such embodiments, the second filter may have a pore size smaller than the filter **222** and provide filtration of gases flowed through the gas supply line **114**, while the filter **222** provides a hydrophobic barrier to prevent leakage of liquid therein.

(29) FIGS. 3A-3C illustrate another valve assembly **300**, which functions in substantially the same manner as the valve assembly **200** depicted in FIGS. 2A-2C, but with a different structure.

Accordingly, FIGS. 3A-3C are described together for clarity, and parts of the valve assembly **300** corresponding to the above-described parts of the valve assembly **200** are marked with the same reference numerals.

(30) As shown, the upper body **232** of the valve assembly **300** includes a base **338**, which generally has a plate-like shape and further includes one or more ridges (e.g., ribs or grooves) **342** extending from a lower surface thereof that form one or more channels within a cavity **340** (shown in FIG. 3C). In certain embodiments, the ridges **342** are annular or semi-annular ridges that circumscribe the distal end of the conduit **236**. The ridges **342** provide added support for the filter **222** when the valve assembly **300** is in an assembled state.

(31) Similar to the cavity **240** of the valve assembly **200**, the cavity **340** fluidly couples with the conduit **236** and has one or more cross-sectional dimensions greater than a width or diameter of the conduit **236**. However, unlike the cavity **240**, the cavity **340** has a cross-sectional area that steeply or abruptly increases from an end of the cavity **340** nearest the arm **234** to an end of the cavity **340**

furthest from the arm **234**. As described above, the increased cross-sectional area of the cavity **340** enables utilization of a larger-area filter **222**, which provides more surface area through which gases may be vented or purged from liquids flowing between the liquid supply line **116** and the infusion line **108**.

(32) The lower body **202** of the valve assembly **300** includes a basin **350** coupled to a flow-through member **358**. The basin **350** is configured to interface and engage with the base **338** of the upper body **232** and secure the filter **222** therebetween. In certain embodiments, the basin **350** couples to the upper body **232** at a lower surface of the base **338** such that the cavity **340** of the base **338** faces a cavity **354** of the basin **350**. At least a portion of the cavity **354** may have a cross-sectional area substantially the same or greater than the cross-sectional area of the end of the cavity **340** nearest the basin **350** so as not to constrict air flow between the upper body **232** and the lower body **202** and vice versa. For example, an end of the cavity **354** opposite the flow-through member **358** may have a cross-sectional area substantially the same or greater than the cross-sectional area of the end of the cavity **340** nearest the basin **350**.

(33) Similar to the base **338**, the basin **350** includes one or more ridges **352** extending from an upper surface thereof into a cavity **354**. The ridges **352** are configured to provide support to the filter **222** when the valve assembly **300** is in an assembled state and may be annular or semi-annular in shape, defining one or more channels therein. In certain embodiments, the ridges **352** circumscribe a proximal end of a channel **356** that fluidly couples the cavity **354** with an intermediate conduit **360** of the flow-through member **358**. The intermediate conduit **360**, in turn, extends and fluidly connects the extensions **206** and **210** of the lower body **202**, which are configured to couple with the liquid supply line **116** and the infusion line **108** at ports **209** and **211**, respectively.

(34) During operation of the valve assembly, infusion liquid from the liquid source may flow through the liquid supply line **116**, into the flow-through member **358** of the lower body **202**, and through the infusion line **108** toward the patient's eye and vice versa. Alternatively, infusion gases from the gas source may flow through the gas supply line **114**, into the upper body **232**, past the filter **222** into the lower body **202**, and through the infusion line **108** toward the patient's eye and vice-versa. Similar to the valve assembly **200**, the disposition of the hydrophobic filter **222** between the upper body **232** and the lower body **202** passively prevents the flow of liquids into the upper body **232** and the gas supply line **114**, while allowing gases to pass therethrough. Thus, like the valve assembly **200**, the valve assembly **300** facilitates venting, purging, and/or back-flow of gases during fluid infusion procedures while preventing the escape of liquid into the gas supply line **114**.

(35) Please note that, as discussed above with reference to the valve assembly **200**, although a single filter **222** is depicted in FIGS. 3A-3C, it is further contemplated that the valve assembly **300** may include two or more filters arranged in a linear or stacked configuration. The two or more filters may be formed of different materials and/or have different pore sizes relative to each other. For example, in certain embodiments, a second filter having a finer pore size may be disposed upstream of the filter **222** to provide additional filtration of gases flowed through the gas supply line **114**, while the filter **222** provides a hydrophobic barrier and prevent liquids from flowing therein.

(36) FIGS. 4A-4C schematically illustrate operational modes of the valve assemblies **200**, **300** during fluid infusion procedures. In particular, FIGS. 4A-4C illustrate the flow of liquid solutions (e.g., BSS), represented by lines **410**, and the flow of gases (e.g., air), represented by lines **420**, through the fluid infusion system **110** having the valve assembly **200**, as described above. Please note that although the valve assembly **200** is depicted in FIGS. 4A-4C, the valve assembly **300** may be utilized in substantially the same manner. Further, please note that unbroken lines (e.g., continuous lines) represent open or active flow, while broken lines (e.g., dashed lines) represent closed or no flow.

(37) FIG. 4A depicts the fluid infusion system **110** during a first operation of liquid infusion, which may be selected and/or controlled by a user (e.g., a surgeon) via a surgical console, such as surgical console **100**. As shown, infusion liquid **410** is controllably flowed between the liquid source **470** and eye **402** via liquid supply line **116**, valve assembly **200**, and infusion line **108**, while air or gas flow through gas supply line **114** is stopped or shut off. To control a pressure within the fluid infusion system **110** and thus, the eye **402**, the user may adjust the direction and flow rate of the liquid **410** to or from the liquid source **470** via the surgical console **100**. The valve assembly **200** (e.g., a flow control valve) enables liquid **410** to flow between the liquid supply line **116** and the infusion line **108**, while also preventing the liquid **410** from flowing into the gas supply line **114** and towards the gas source **480** due to the presence of the hydrophobic filter **222**. Accordingly, the valve assembly **200** provides a passive means of preventing leakage of liquid **410** into gas supply line **114**, which contrasts with conventional flow control valves that may allow the escape of at least some liquid **410** into the gas supply line **114** during use thereof.

(38) FIG. 4B depicts the fluid infusion system **110** during a second operation of liquid infusion in which the pressure of air **420** within the gas supply line **114** is actively modulated while infusion liquid **410** is flowed between the liquid source **470** and eye **402**. As described above, the pressure within the fluid infusion system **110** and the eye **402** is controlled by adjusting the direction and flow rate of the liquid **410** to or from the liquid source **470** via the surgical console **100**. When left unchecked, pressure within the gas supply line **114** may inadvertently build up during infusion and cause air **420** to leak into the liquid **410** being injected into the eye **402**, thereby negatively affecting the intraocular pressure thereof. Therefore, in certain embodiments, it may be desired to apply a vacuum pressure (e.g., negative pressure) to the gas supply line **114** to vent the gas supply line **114** and prevent the undesired escape of air **420** into the liquid **410** as bubbles. In certain embodiments, active venting of the gas supply line **114** may also be desired to purge the infusion liquid **410** of gases already trapped therein as the liquid **410** passes into the infusion line **108**.

(39) Since conventional flow control valves cannot prevent the leakage of liquid **410** into the gas supply line **114**, venting of the gas supply line **114** with a conventional valve is extraordinarily difficult. In comparison, as a result of the hydrophobic filter **222**, the valve assembly **200** facilitates active venting of the gas supply line **114** during infusion of liquid **410** into the eye **402**, thus reducing or eliminating the possibility of unwanted gases being flowed into eye **402** and disrupting the intraocular pressure therein.

(40) FIG. 4B is further representative of the fluid infusion system **110** during an infusion fluid back-flow operation. Back-flow of infusion fluids may be necessitated when the eye **402** is injected, via a separate cannula or injection device, with a retinal tamponade (or other fluid treatment) such as intraocular air/gas, silicone oil, or perfluoron. As a result, infusion fluids previously flowed through the infusion line **108** may need to be back-flowed. Because conventional flow control valves cannot backflow or purge gases into the gas supply line **114** without leakage of infusion liquid, only a limited volume of infusion fluids can be back-flowed without risking the chance of liquid leakage into the gas supply line **114** or gas leakage into the liquid supply line **116**. In contrast, the hydrophobic filter **222** of the valve assembly **200** in FIG. 4B enables backflow of gases into the gas supply line **114** without leakage of infusion liquids, thus allowing a greater volume of the infusion fluids to be back-flowed into their respective supply lines and further enabling a greater volume of treatment fluids to be injected into the eye **402**.

(41) FIG. 4C depicts the fluid infusion system **110** during a third operation of liquid infusion. The operational mode depicted in FIG. 4C may be performed, for example, during a fluid-air exchange to help push out subretinal fluid from the intraocular space of the eye **402**. As shown, air **420** is flowed from the gas source **480** to the eye **402**, while liquid flow through the liquid supply line **116** is shut off to prevent escape of liquid **410** into the infused air **420**. Accordingly, the pressure within the fluid infusion system **110** and the eye **402** is controlled by adjusting the direction and flow rate of the air **420** to or from the gas source **480** via the surgical console **100**.

(42) In summary, embodiments of the present disclosure include structures and mechanisms for improved intraocular pressure maintenance during ophthalmic procedures, and in particular, improved fluid control valves for intraocular fluid infusion. The valve assemblies described above include embodiments wherein a hydrophobic filter is disposed between a gas supply line and a liquid supply line and/or infusion line. The utilization of the hydrophobic filter enables bi-directional flow of gases between a gas supply line and the patient's eye, while also passively preventing the leakage of liquids into the gas supply line. Accordingly, the aforementioned valve assemblies are particularly beneficial during fluid infusion of the intraocular space, as gas may be vented from infusion liquids during infusion or back-flowed from the eye during injection of other treatments, thus allowing better control of the intraocular pressure within the eye.

(43) While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A valve assembly configured for bi-directional flow of gas and infusion liquid therethrough during ophthalmic procedures, comprising: a first portion configured to allow a bidirectional flow of gas therethrough, comprising: a first conduit having a first port configured to couple to a source of gas; and a first cavity having a first end and a second end opposite the first end, the first end fluidly coupled to the first port via the first conduit, the second end having a cross-sectional area greater than a cross-sectional area of the first conduit; a second portion configured to allow bi-directional flow of gas and infusion liquid therethrough, comprising: a second conduit having a second port and a third port on an opposing side of the second conduit as the second port, wherein the second port, the second conduit, and the third port provide a first flow passageway through the second port, the second conduit, and the third port; and a second cavity in fluid communication with the second port and the third port, the second cavity adjacent to the first cavity; wherein the second port is configured to couple to a source of liquid and wherein the third port is configured to couple to an infusion line for an ophthalmic surgical site; and a filter partitioning the first cavity from the second cavity, the filter comprising a hydrophobic membrane partially defining the second cavity along a planar face of the hydrophobic membrane; wherein the first portion, the filter, and the second portion are configured to allow a flow of gas from the first port, through the first portion, through the filter, through the second portion, and out of the third port; wherein the first portion, the filter, and the second portion are further configured to allow a flow of gas from the third port, through the second portion, through the filter, through the first portion, and out the first port; and wherein the first portion, the filter, and the second portion are further configured to allow a flow of infusion liquid from the second port, through the second portion, and out the third port while further configured to provide vacuum from the first port to aspirate gas from the flow of infusion liquid in the second portion, such that the aspirated gas flows from the infusion liquid in the second portion, through the filter, through the first portion, and out the first port.
2. The valve assembly of claim 1, wherein the hydrophobic membrane is configured to prevent a flow of liquid from the second cavity into the first cavity while allowing bi-directional flow of gas therebetween.
3. The valve assembly of claim 1, wherein the hydrophobic membrane comprises at least one of polytetrafluoroethylene (PTFE), expanded PTFE (ePTFE), polycarbonate track etch (PCTE), polyester, nylon, cellulose, cellulose nitrate (CN), cellulose acetate (CA), polyethersulfone (PES), glass fiber, and acrylic copolymer.
4. The valve assembly of claim 3, wherein the filter further comprises a backing coupled to the hydrophobic membrane, the backing comprising at least one of polyester, polyethylene, polypropylene, and nylon.

5. The valve assembly of claim 1, wherein the hydrophobic membrane has a pore size in a range of 0.1 μm to 10.0 μm and a thickness in a range of 150 μm to 300 μm .
6. The valve assembly of claim 1, wherein the first cavity further comprises one or more semi-annular or annular ridges disposed therein, the one or more ridges defining one or more channels.
7. The valve assembly of claim 6, wherein the second cavity further comprises one or more semi-annular or annular ridges disposed therein, the one or more ridges defining one or more channels.
8. The valve assembly of claim 1, wherein at least a portion of the second cavity has a cross-sectional area substantially the same or greater than the cross-sectional area of the second end of the first cavity.
9. The valve assembly of claim 1, wherein a second flow passageway between the first conduit and the first cavity is perpendicular to the first flow passageway through the second port, the second conduit, and the third port.
10. The valve assembly of claim 1, wherein the third port comprises a luer lock.
11. The valve assembly of claim 1, wherein the first flow passageway between the second cavity and the second conduit has a smaller diameter than a second flow passageway between the first cavity and the first conduit.
12. The valve assembly of claim 1, wherein a second flow passageway through the first port is parallel to the first flow passageway through the second port, the second conduit, and the third port.
13. The valve assembly of claim 1, wherein the flow passageway through the second port, second conduit, and third port is straight.
14. The valve assembly of claim 13, wherein the first flow passageway through the second port, the second conduit, and the third port is parallel to the planar face of the hydrophobic membrane.
15. The valve assembly of claim 13, wherein the first flow passageway through the second portion provides unimpeded flow from the second port to the third port without an intervening filter.
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